



Introduction to Financial Markets

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Session 1



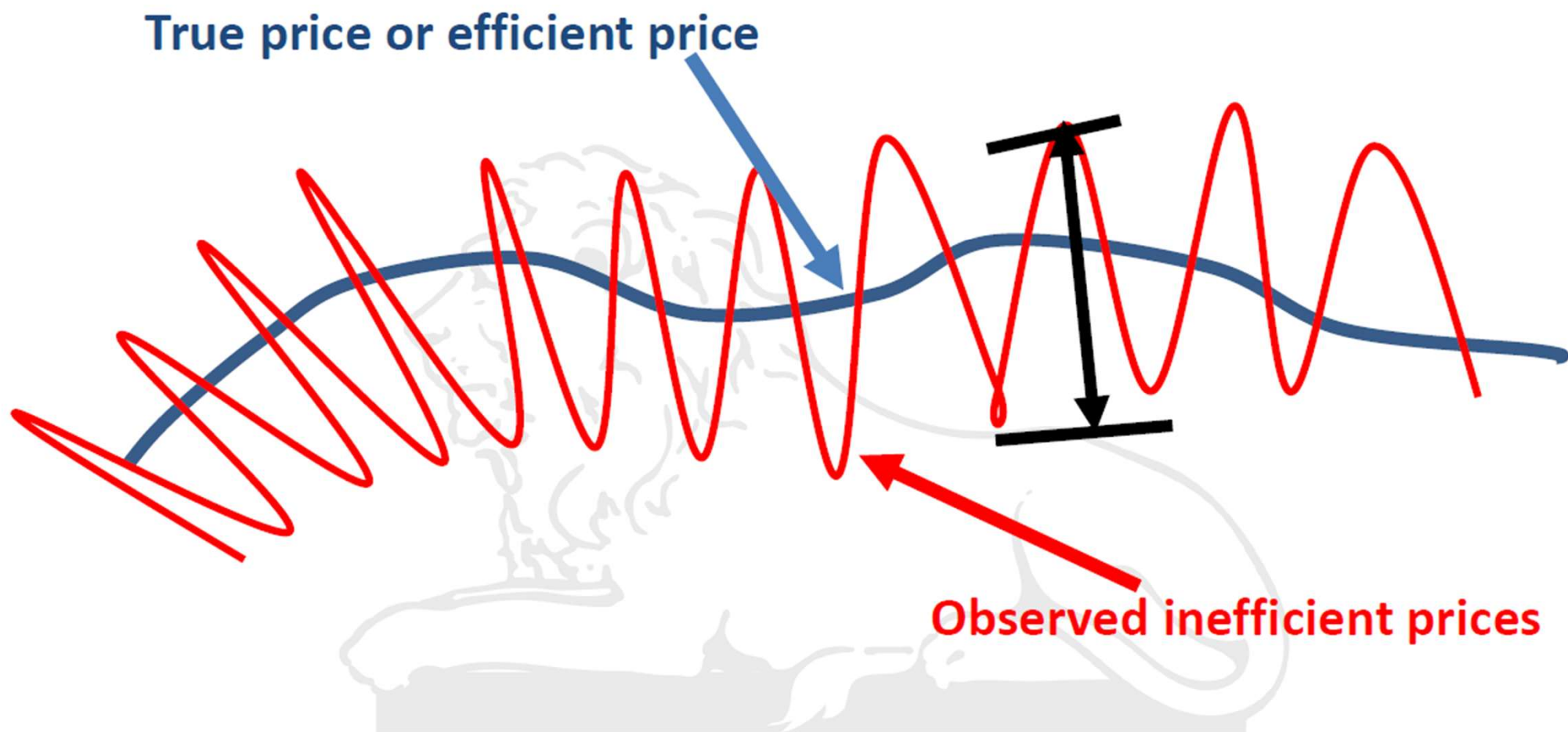
Market Efficiency



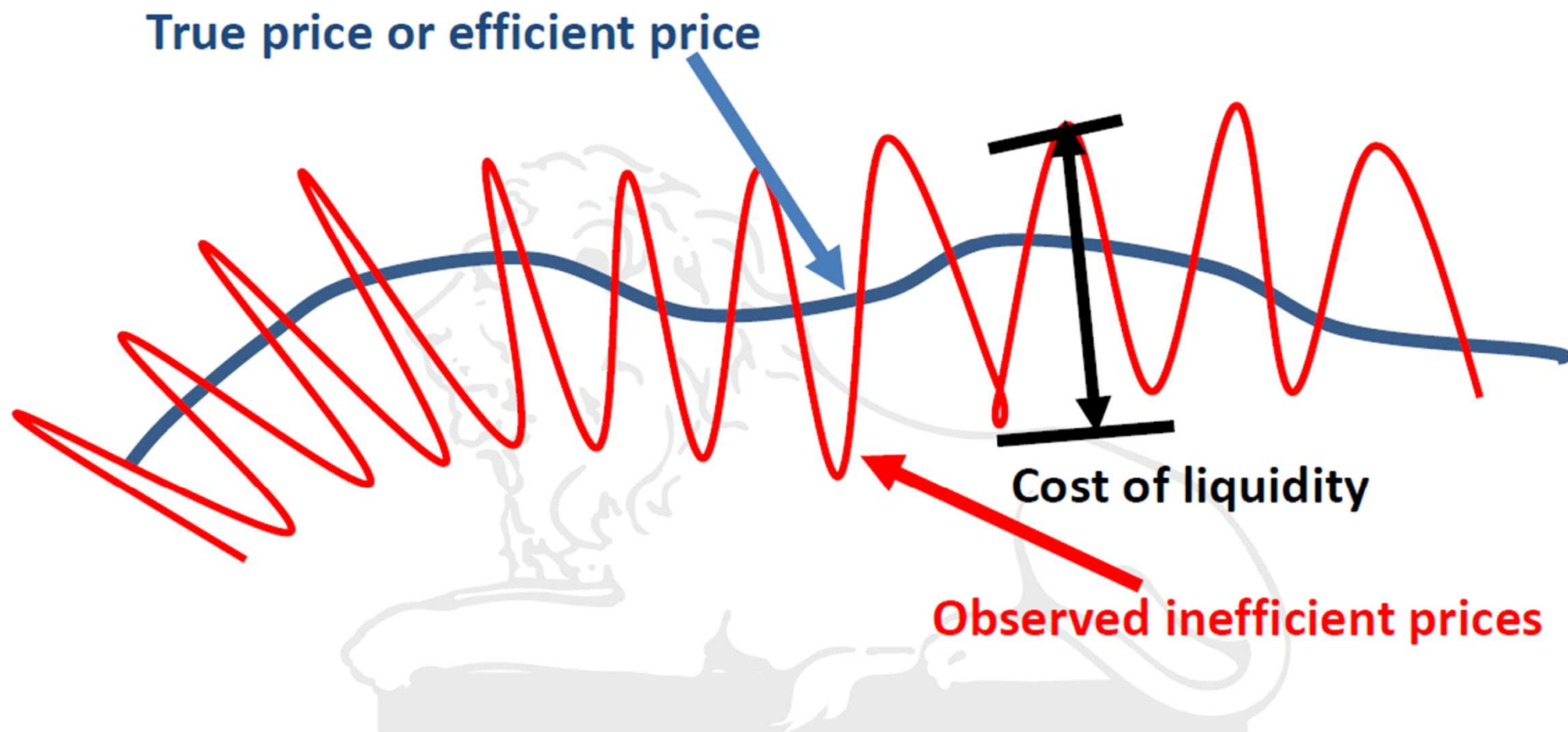
Introduction: Why do we need financial markets

- To channel the savings of an economy towards productive and efficient sectors: **Financial Intermediation**
- Banks conventionally play this role
- However, financial markets allow investors to directly participate in the process of choosing the best sector for himself
- Thus, less bureaucratic hassles or principal-agent problem
- To this end, an economy needs efficient and liquid markets
- **What is efficient and liquid ?**

Market efficiency



Market efficiency



Market efficiency

- The true price or efficient price or fundamental price is called so because it reflects all the “fundamental information” about security
- The information that affects the cash flows of the firm(e.g., macroeconomic information, or information about the firm)
- However, this price is not observed
- What we observe on stock markets is a mix of fundamental prices and noise
- $P_{observed} = P_{true} + e_{noise}$



Market efficiency

- This noise, though a small component, is at times sufficient for informed traders to conduct **arbitrage trading (Speculation?)**
- **Riskless Arbitrage trading:** Same stock is traded in different markets (BSE vs NSE, Cash vs Future). If price in one of the markets deviates too much from its true value, informed traders start selling/buying the stock in that market and take opposite position in the other market (buying/selling). Since they are not taking any risk, this is called **riskless arbitrage** (In practical life, no arbitrage is riskless)

Market efficiency

- However, for arbitrage to take place the price has to move sufficiently to cover the trading costs (costs of liquidity)
- Therefore, the forces of arbitrage only work when the price moves out of the cost of liquidity window
- Then forces of arbitrage kick-in and pull the prices towards the efficient price until it moves back in the liquidity window
- Thus, price keeps oscillating within the cost of liquidity window
- **What is the amplitude of this window?**

Market efficiency

- This amplitude reflects cost of liquidity
- At times prices move-beyond this amplitude and exhibit high fluctuations
- Then we say that prices are volatile
- This happens during some crisis periods or negative news
- A higher volatility means that prices divert away from their true values
- Different markets exhibit different levels of volatility depending upon the level of efficiency

Market efficiency

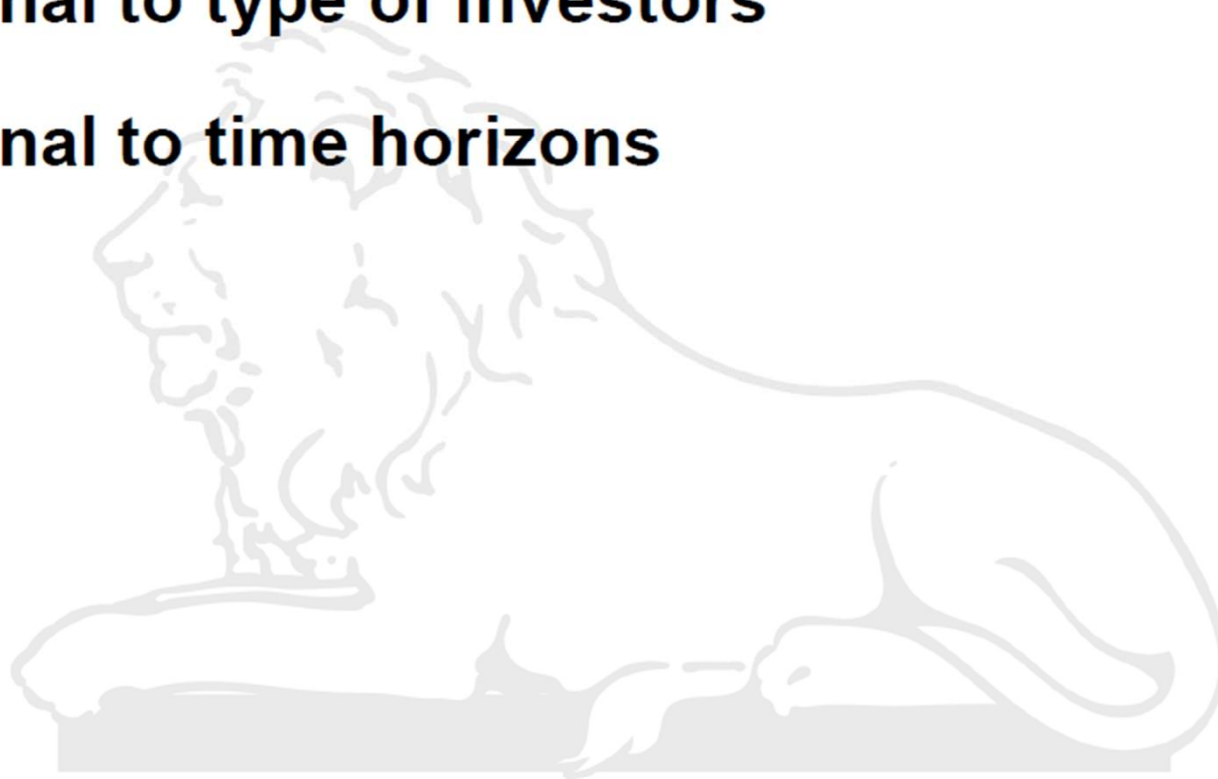
- **Weak-form efficiency:** When a trader can not make profit using the information contained in past prices, markets are said to be weak-form efficient
- **Semi-strong form efficiency :** When a trader can not make profit using the information contained in past prices and public
- **Strong form efficiency :** When a trader can not make profit using the information contained in past prices, public information, as well as private information

Market efficiency

- **Not efficient at all:** Make technical trading strategies, using the information from past price and volume
- **Weak-form efficiency:** Make profits by collecting public information
- **Semi-strong form efficiency :** Make profits if one has private information
- **Strong form efficiency :** Can not beat the market, so follow the market by investing in indices that reflect the broad market movement (Nifty-50)

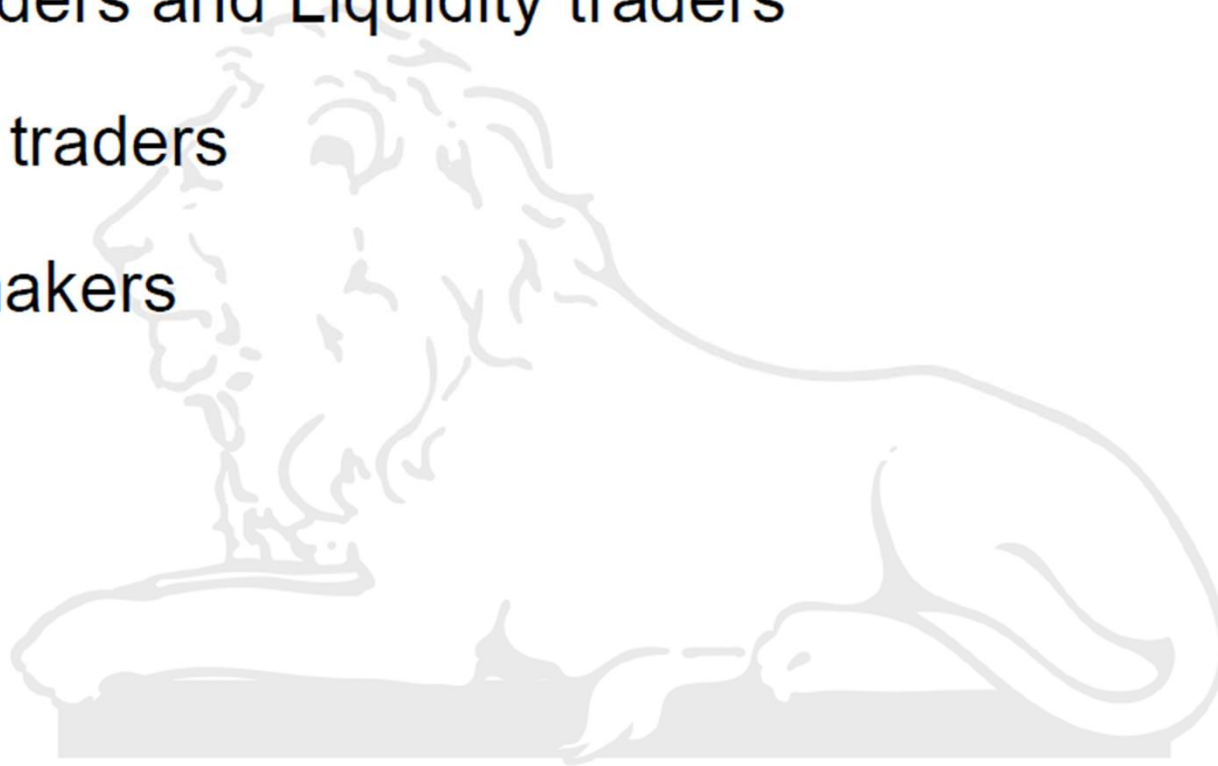
Market efficiency

- Are these definitions absolute?
- Conditional to type of investors
- Conditional to time horizons



Few other definitions

- **Types of traders**
- Noise traders and Liquidity traders
- Informed traders
- Market makers



Expectations



Expectations Operator 'E'

- Any **random probabilistic variable** (e.g., **Coin toss**) is often represented through expectations operator: $E(\text{Profit from coin toss})$
- Any random variable attains multiple values. For example, a coin-toss can obtain two values with 50% odds for any outcome (Heads win 100; Tails loss 50)
- **$E(\text{Profit from coin toss}) = 100 \cdot 0.5 + (-50) \cdot 0.5 = 25$**

Expectations Operator 'E'

- For example, if there are 'n' possibilities of an event, $y_1, y_2, y_3, \dots, y_n$ each with possibilities $p_1, p_2, p_3, p_4 \dots p_n$, then expectations operator is defined as

- $E(y) = p_1 * y_1 + p_2 * y_2 + p_3 * y_3 + \dots + p_n * y_n$

- This is also called probability weighted mean
- If all the probabilities are assumed to be equal then

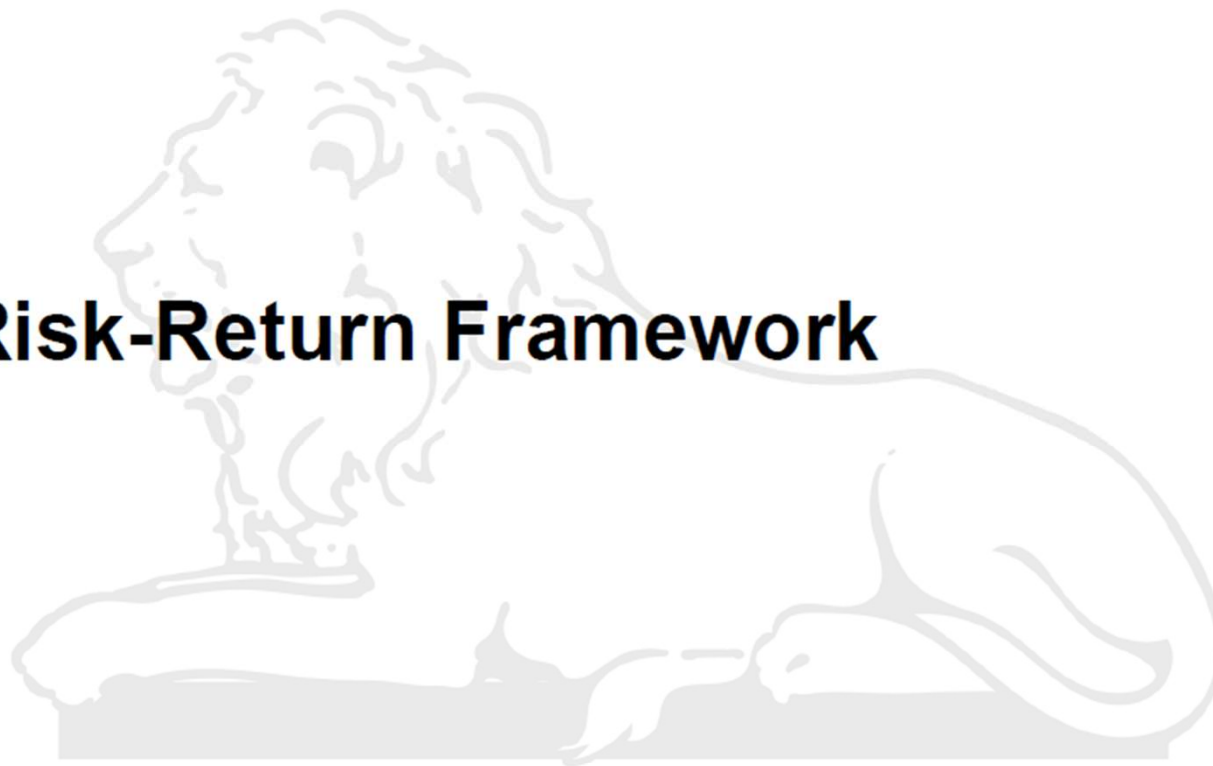
$$p_1 = p_2 = \dots = p_n = \frac{1}{n}$$

- Then $E(y) = \frac{1}{n} (y_1 + y_2 + y_3 + \dots + y_n)$, i.e., simple average of Y's

Expectations Operator 'E'

- $E(\text{Profit from coin toss}) = 100 \cdot 0.5 + (-50) \cdot 0.5 = 25$
- **Risk Neutral:** Equally happy (utility) with the gamble with expectation of Rs 25 or the certain Rs 25
- **Risk Averse:** More happy (utility) with the certain Rs 25 as compared to a gamble with expectation of Rs 25
- **Risk Loving:** Less happy (utility) with the certain Rs 25 as compared to a gamble with expectation of Rs 25

Back to Risk-Return Framework





Returns: computation of average actual returns

- This takes us to a very important concept of expected returns
- Different from actual returns: builds expectation of future
- Consider an SBI FD and investment into mutual funds. How much return we should expect from these instruments. From SBI FD probably a fixed return of say 7%. What about investment into mutual funds? Maybe 10%, or even higher. Why should we expect different returns from these instruments?



Returns: expected returns

- Different from actual returns: builds expectation of future
- Consider an SBI FD and investment into mutual funds. How much return we should expect from these instruments. From SBI FD probably a fixed return of say 7%. What about investment into mutual funds? Maybe 10%, or even higher. Why should we expect different returns from these instruments? Because these instruments offer different degrees of risk to the owners of the instruments. For example, SBI-FD is safe. And for a very high degree of surety you know that your principal and interest amounts will be paid back at the time instrument matures, i.e., the holding period. However, in case of mutual funds, there is no such surety, and you may even get an amount that is lesser than what you invested. This uncertainty of outcome is called risk.

Risk in Financial Markets



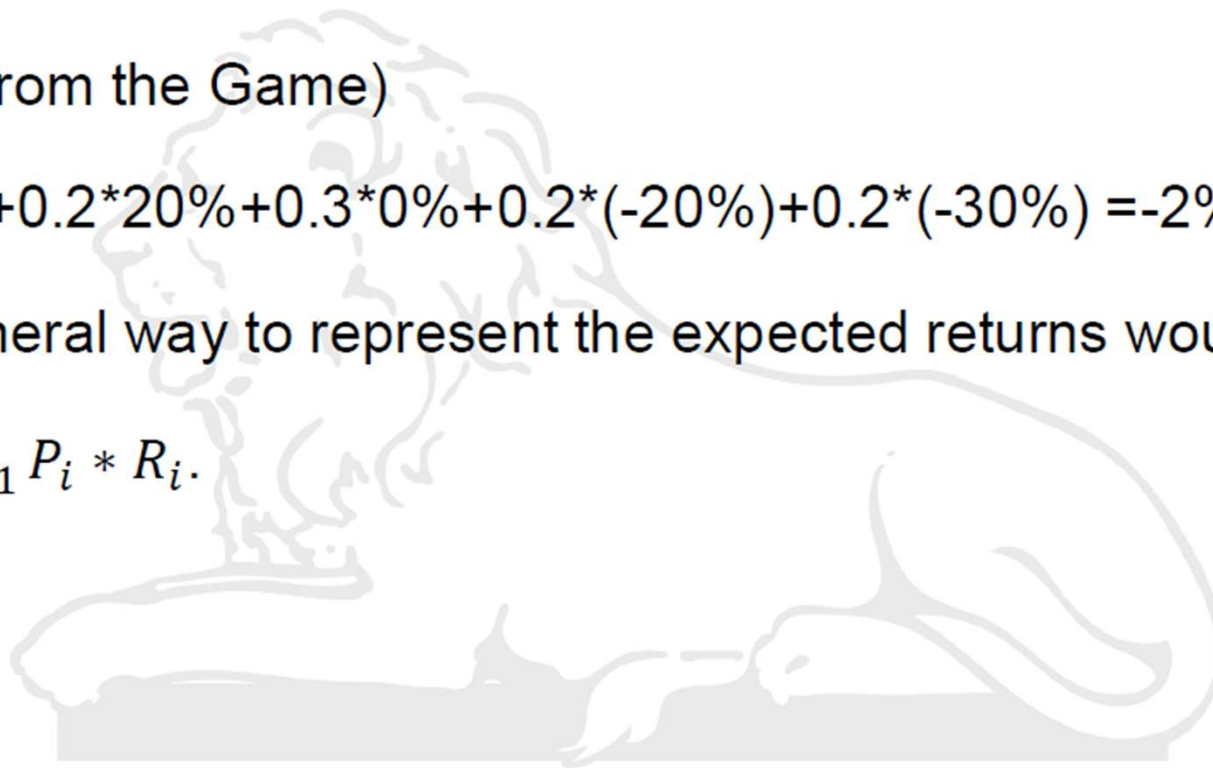
Returns: expected returns

- A more general way to represent the expected returns would be like this
- $E(R_{i,t}) = \sum_{i=1}^{N=5} P_i * R_i.$

S. No. (t)	Probabilities (P_i)	Returns
1	0.1	40%
2	0.2	20%
3	0.3	0%
4	0.2	-20%
5	0.2	-30%

Returns: expected returns

- Different from actual returns: builds expectation of future
- What is our expected outcome in this game?
- $E(\text{Return from the Game})$
- $= 0.1 \cdot 40\% + 0.2 \cdot 20\% + 0.3 \cdot 0\% + 0.2 \cdot (-20\%) + 0.2 \cdot (-30\%) = -2\%$
- A more general way to represent the expected returns would be like this
- $E(R_i) = \sum_{i=1}^N P_i * R_i.$



Returns: expected returns

- A more general way to represent the expected returns would be like this
- $E(R_t) = \sum_{t=1}^T P_t * R_t$.
- What if we don't have these probabilities. We only have past returns and all returns are equally likely
- i.e., $P_1 = P_2 = P_3 = P_4 \dots \dots = P_T$. And since we know that $\sum_{t=1}^T P_t = 1$.

Therefore, $P_1 = P_2 = P_3 = P_4 \dots \dots = P_T = \frac{1}{T}$

- $E(R_t) = \overline{R_t} = \frac{1}{T} \sum_{t=1}^T R_t$

Returns: expected returns

- Different from actual returns: builds expectation of future
- A more general way to represent the expected returns would be like this
- $E(R_{i,t}) = \sum_{t=1}^T P_i * R_{i,t} = (1/5) * (40\% + 20\% + 0\% - 20\% - 30\%)$

S. No. (t)	Expected Returns
1	40%
2	20%
3	0%
4	-20%
5	-30%

- But what are these expectations based upon?
- Risk

Risk: what is risk and how to measure it?

- Risk in financial markets
- Uncertain outcomes lead to risk. If the outcome is certain (SBI FD) then then there is no risk. Risk-free assets
- **Which one is more risky (1) toss a coin; (2) through a dice; (3) pick a card from the card-deck**
- **A person can be risk neutral; risk averse, risk taking**
- How to measure risk

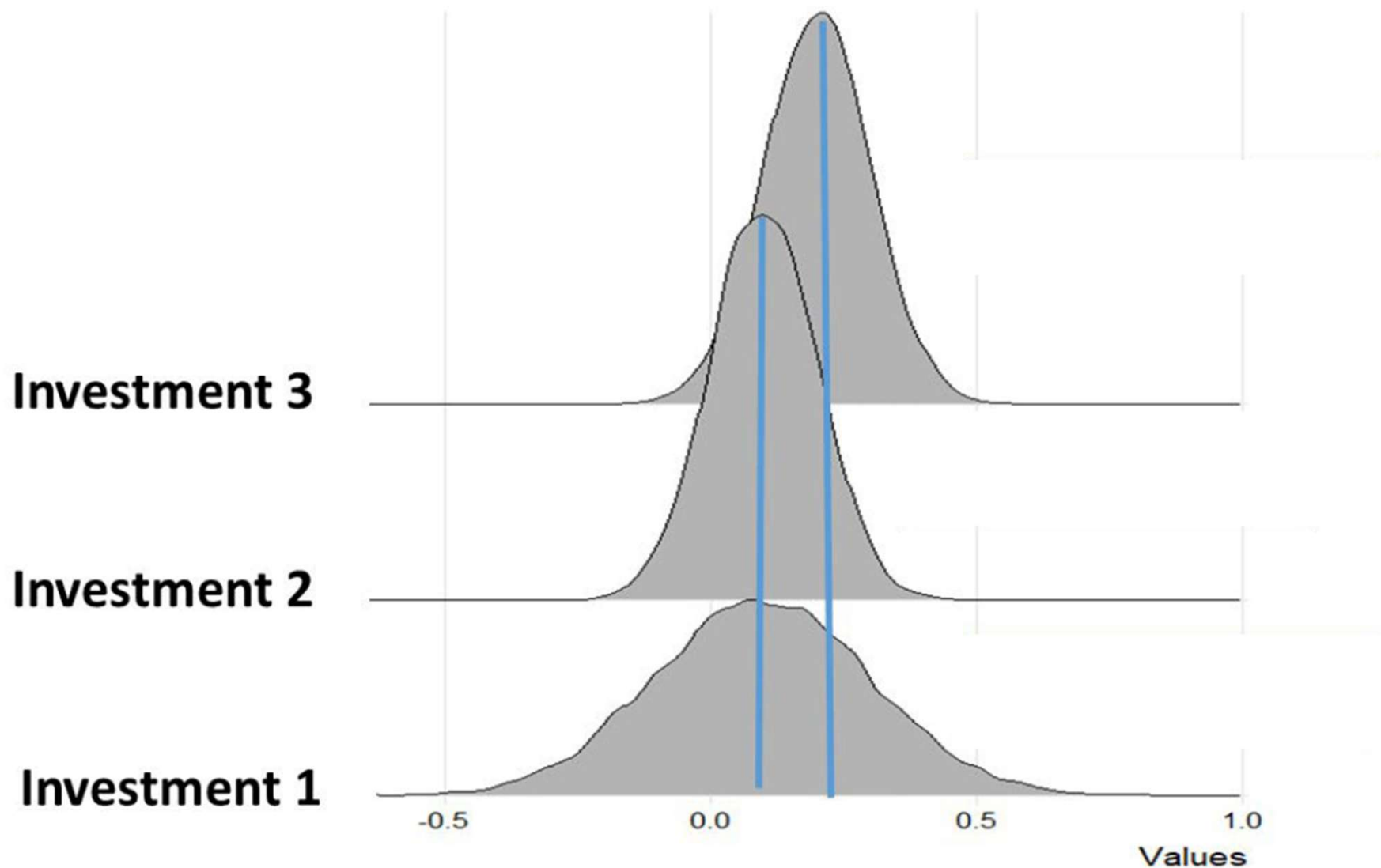
Risk: what is risk and how to measure it?

- How to measure risk: variance (σ_t^2) or standard deviation (σ_t)
- $E(R_t - \bar{R}_t)^2 = \text{Variance } (\sigma_t^2) = \sum_{t=1}^T P_t (R_t - \bar{R}_t)^2$
- Again, for past observations that are equally likely
- i.e., $P_1 = P_2 = P_3 = P_4 \dots \dots = P_T$. And since we know that $\sum_{t=1}^T P_t = 1$. Therefore, $P_1 = P_2 = P_3 = P_4 \dots \dots = P_T = \frac{1}{T}$
- **Variance** $(\bar{\sigma}_t^2) = \frac{1}{T} \sum_{t=1}^T (R_t - \bar{R}_t)^2$
- Often working with samples we use $\bar{\sigma}_t^2 = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R}_t)^2$

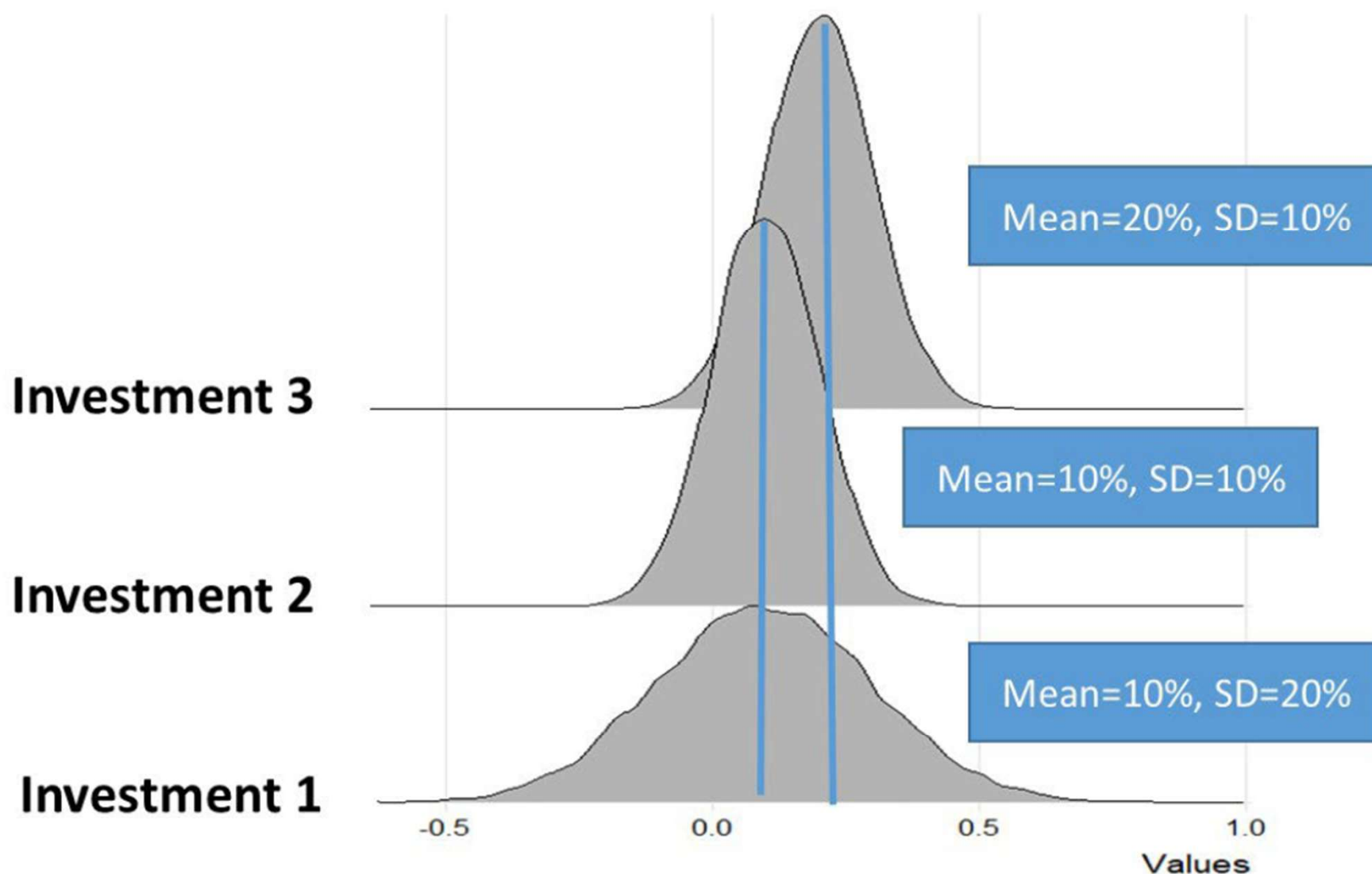
Expectations operator for co-variance

- $\sigma_{12} = E[(R_{1t} - E(R_{1t}))(R_{2t} - E(R_{2t}))] =$
- $= E[(R_{1t} - \bar{R}_1)(R_{2t} - \bar{R}_2)] = \sum_{t=1}^T P_t [(R_{1t} - \bar{R}_1)(R_{2t} - \bar{R}_2)]$
- When all the scenarios are equally likely
- $E[(R_{1t} - \bar{R}_1)(R_{2t} - \bar{R}_2)] = \frac{1}{T} \sum_{t=1}^T [(R_{1t} - \bar{R}_1)(R_{2t} - \bar{R}_2)]$
- $\sigma_{12} = \frac{1}{T} \sum_{t=1}^T [(R_{1t} - \bar{R}_1)(R_{2t} - \bar{R}_2)]$
- **Correlation coefficient** $\rho_{12} = \frac{\sigma_{12}}{\sigma_1 \sigma_2}$

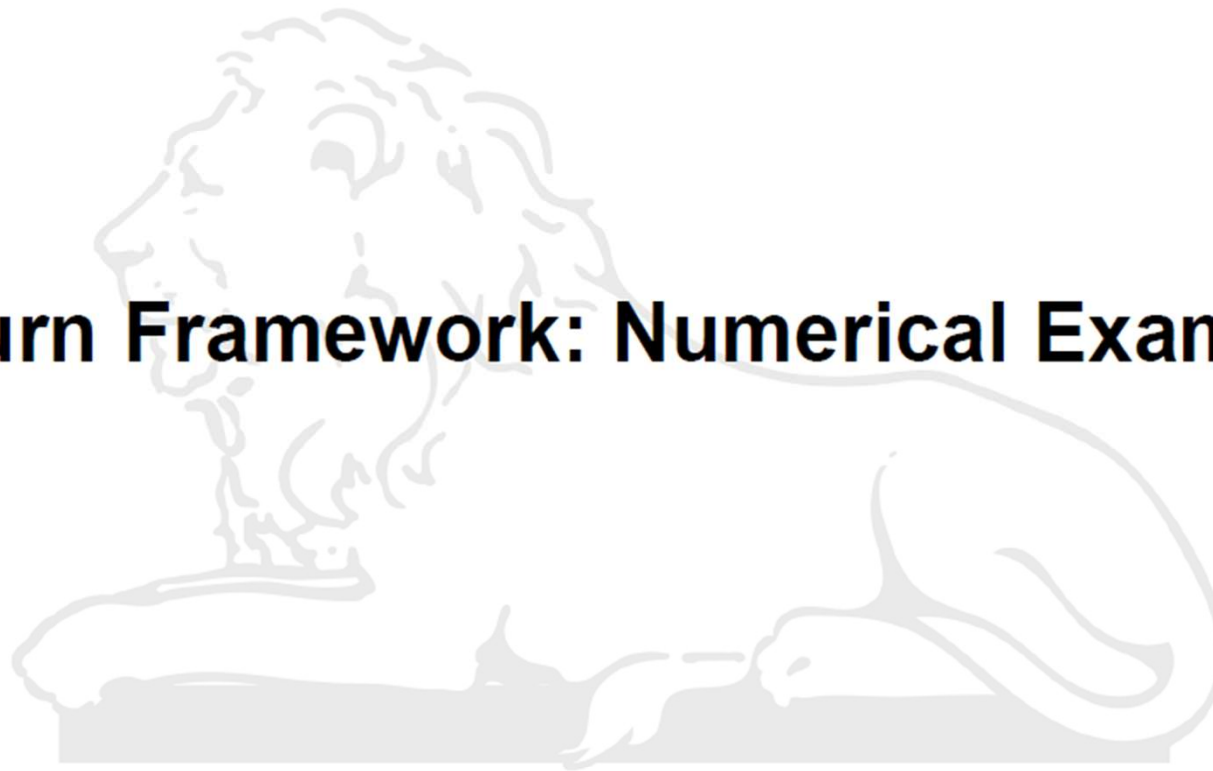
Distribution of returns



Distribution of returns



Risk-Return Framework: Numerical Example



Risk: what is risk and how to measure it?

- Let us go back to our example 'Game' for which we computed expected returns with given probabilities. Now, we will compute the expected variance for the same example using probabilities.
- $E(R_t - \bar{R}_t)^2 = \text{Variance } (\sigma_t^2) = \sum_{t=1}^T P_t (R_t - \bar{R}_t)^2$

S. No. (t)	Probabilities (P_t)	Expected Returns (Mean $\bar{R}_t = -2\%$)
1	0.1	40%
2	0.2	20%
3	0.3	0%
4	0.2	-20%
5	0.2	-30%

Risk: what is risk and how to measure it?

Probabilities (P_t)	Expected Returns (Mean $\bar{R}_t = -2\%$)	Mean deviation ($R_t - \bar{R}_t$)	Squared Deviation ($R_t - \bar{R}_t$) ²	Probability* Squared Deviation $P_t * (R_t - \bar{R}_t)^2$
0.1	40%	42%	0.1764	0.01764
0.2	20%	?	?	?
0.3	0%	?	?	?
0.2	-20%	?	?	?
0.2	-30%	?	?	?

- $\sigma_t^2 = \sum_{t=1}^T P_t (R_t - \bar{R}_t)^2 = 0.01764 + ? + ? + ? + ? = A$
- And the value of standard deviation = $\sqrt{(\sigma_t^2)} = \sqrt{A} = B$

Risk: what is risk and how to measure it?

Probabilities (P_t)	Expected Returns (Mean $\bar{R}_t = -2\%$)	Mean deviation ($R_t - \bar{R}_t$)	Squared Deviation ($R_t - \bar{R}_t$) ²	Probability* Squared Deviation P_t * ($R_t - \bar{R}_t$) ²
0.1	40%	42%	0.1764	0.01764
0.2	20%	22%	0.0484	0.00968
0.3	0%	-2%	0.0004	0.00012
0.2	-20%	-18%	0.0324	0.00648
0.2	-30%	-28%	0.0784	0.01568

- $\sigma_t^2 = \sum_{t=1}^T P_t (R_t - \bar{R}_t)^2 =$
 $0.01764 + 0.00968 + 0.00012 + 0.00648 + 0.01568 = 0.0496$
- And the value of standard deviation = $\sqrt{(\sigma_t^2)} = \sqrt{0.0496}$
 $= 0.2227$ or 22.27%

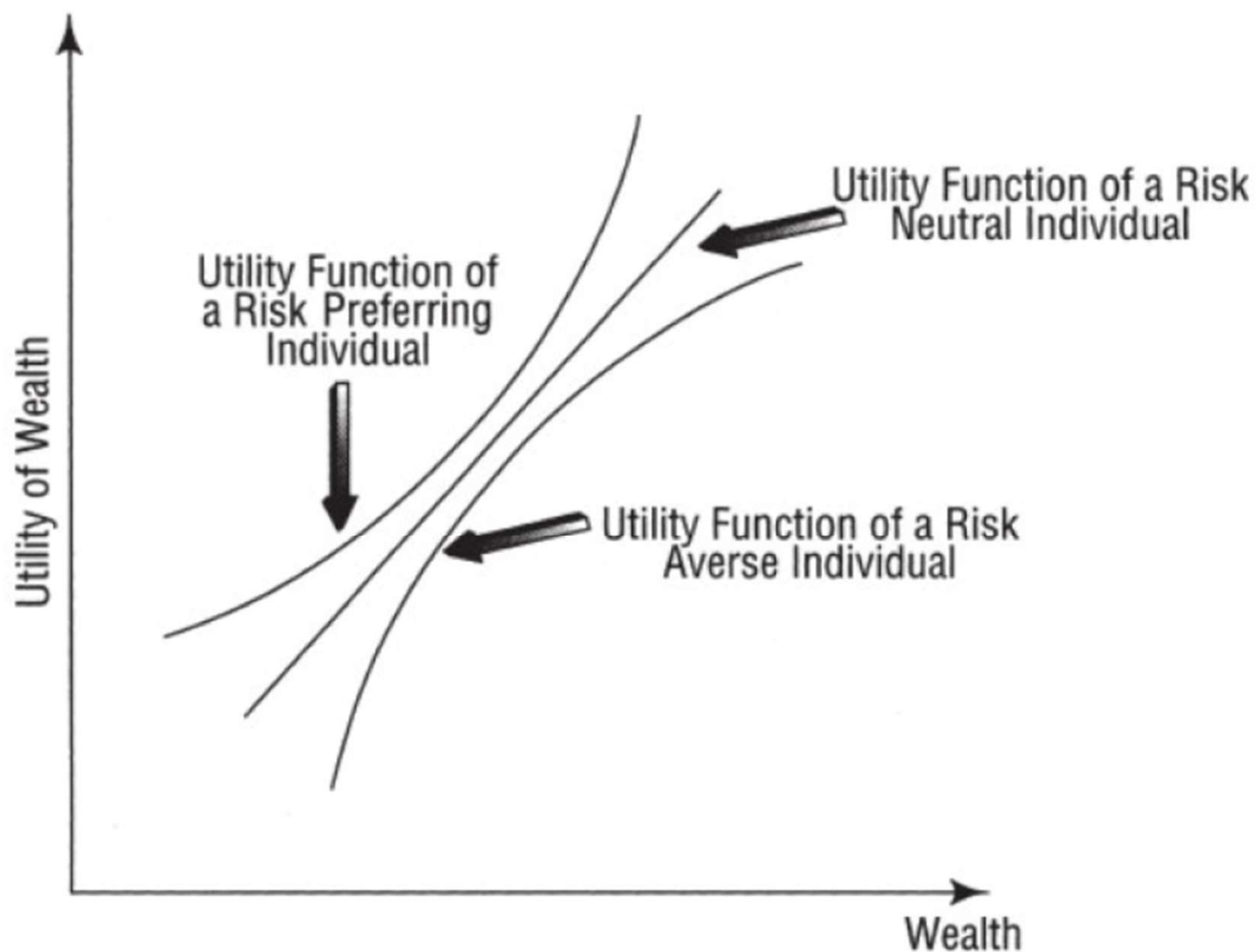
Risk Preferences



Introduction: Risk Preference

- Individuals are described by their utility function. These utility functions describe our preferences for different outcomes
- For a given level wealth W , an individual may have the utility of wealth as $U(W)$. Unlike all normal things, for wealth, it is reasonable to assume $U'(W) > 0$, i.e., more is preferred to less or “**non-satiation**”
- This means that increase in wealth will always lead to increase in utility, no matter how small
- With this, individuals are classified as **(1) Risk neutral; (2) Risk Averse; (3) Risk preferring.**

Introduction: Risk Preference





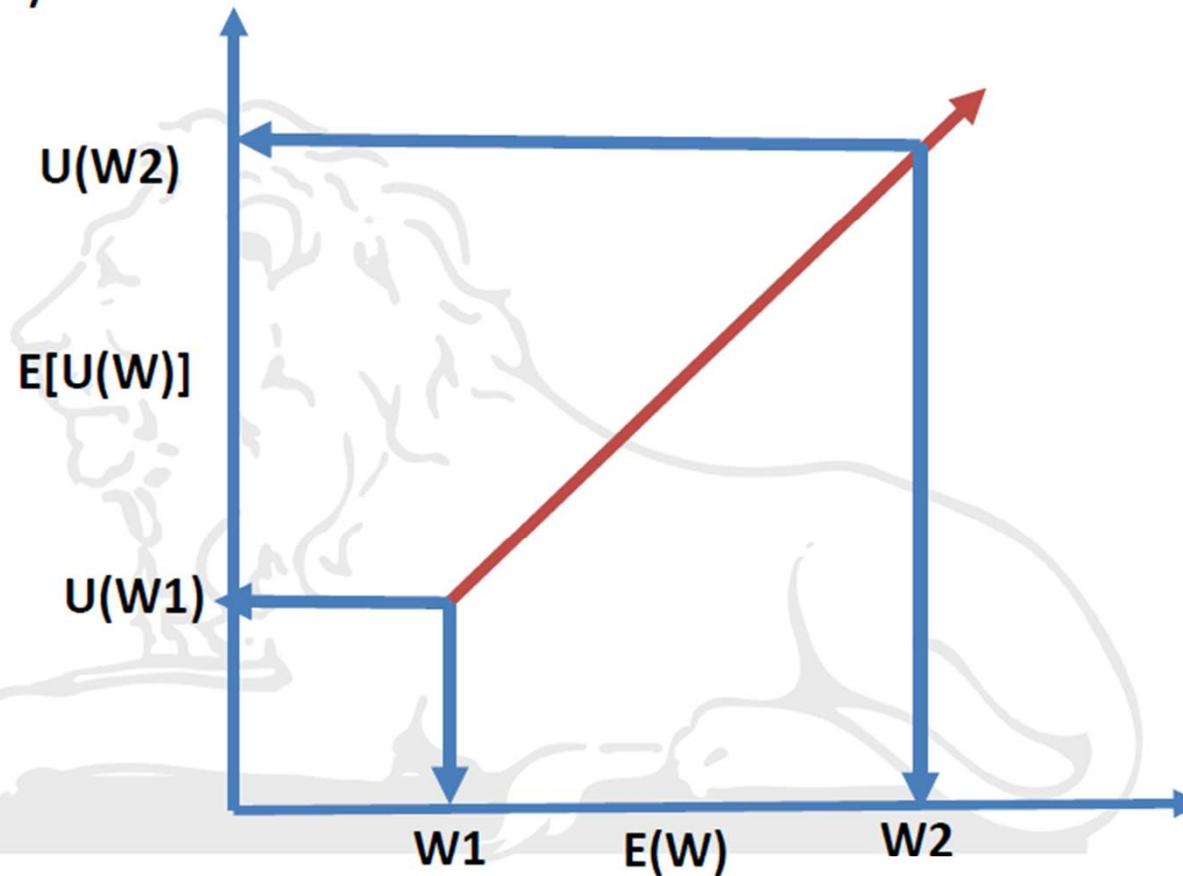
Risk Neutral Person

- If the individual is indifferent in receiving the expected value of gamble
- ***That means, in the previous example, if you make a bet the expected value of winning the bet is Rs 25. If you are indifferent in directly getting Rs 25 vs taking this bet***
- This also means that in this case, expected wealth is important for the investor, not the risk (variance) of the wealth
- Therefore, utility is a linear function of wealth here: $U''(W) = 0$
- This also means that changing the risk of outcome has no effect on the utility (well-being for a given level of wealth): **$U[E(W)] = E[U(W)]$.**

Risk Neutral Person

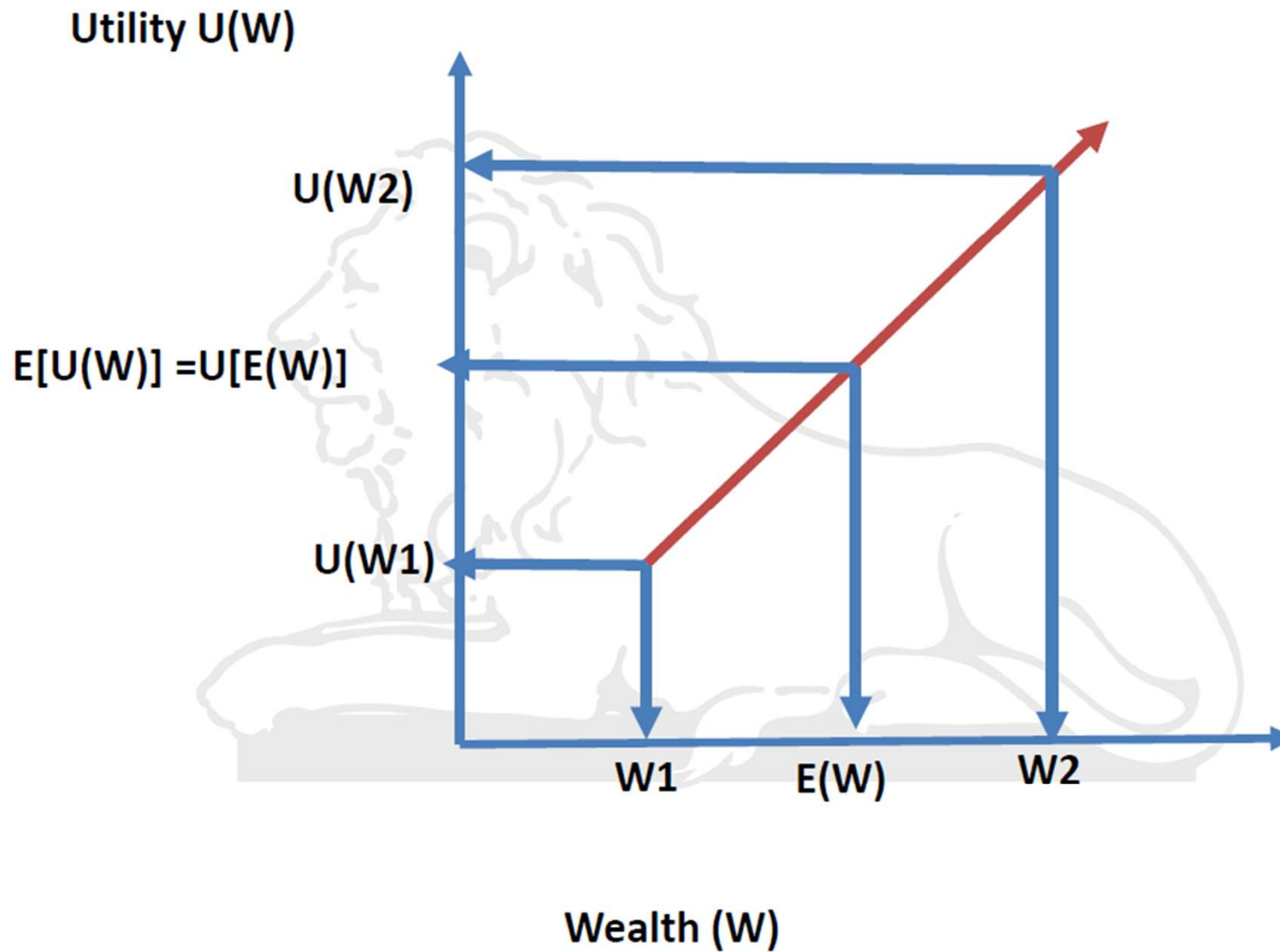


Utility $U(W)$



Wealth (W)

Risk Neutral Person



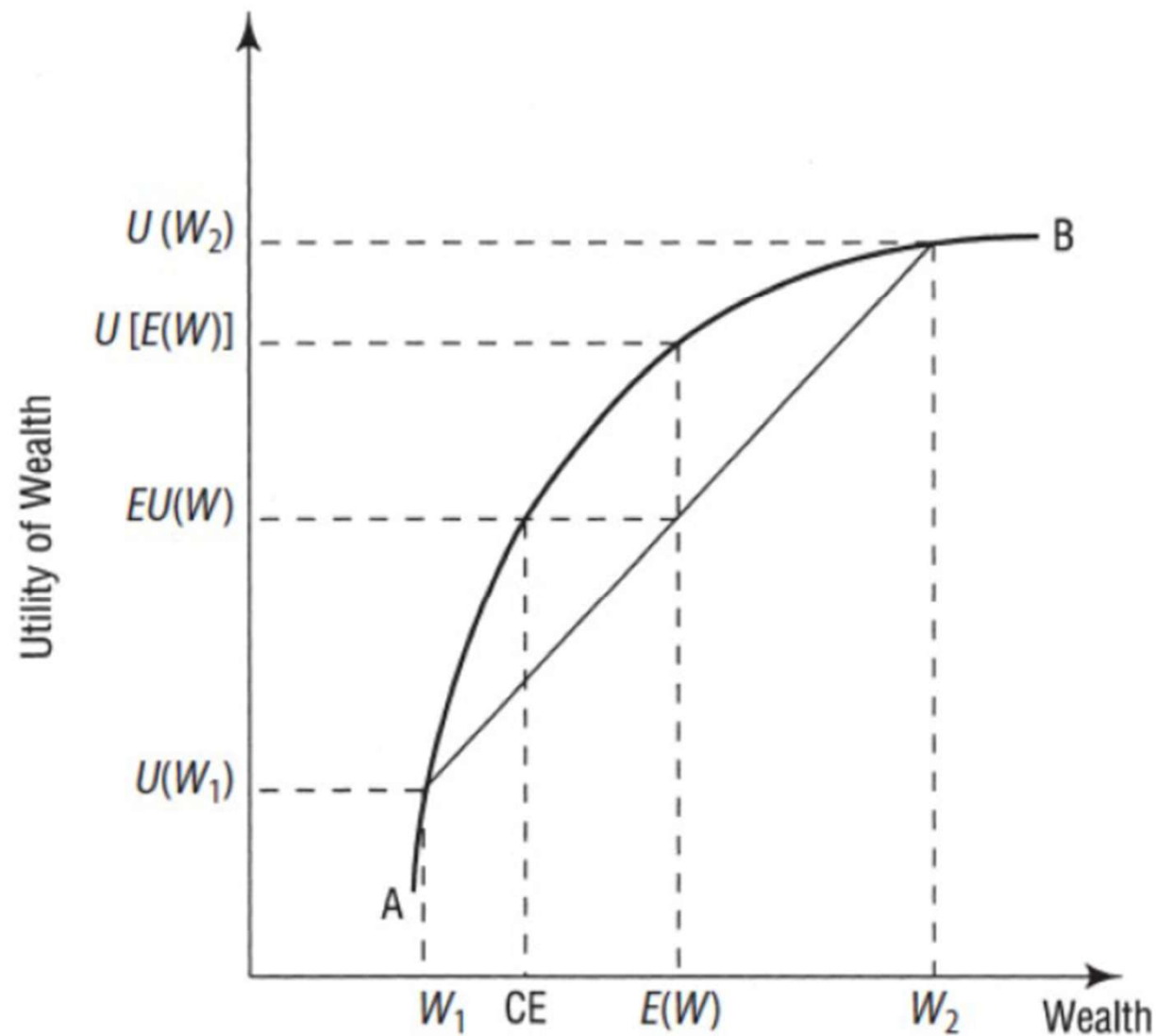
Risk Averse Person

- Utility function of a risk averse individual is concave, that is

$$U''(W) < 0$$

- This also means that the individual prefers a certain (ensured) amount over the bet (gamble) with the same expected value: **$U[E(W)] > E[U(W)]$**
- This also means that risk averse individuals prefer less risk to more, and they demand additional **risk-premia** to take the extra-risk

Risk Averse Person



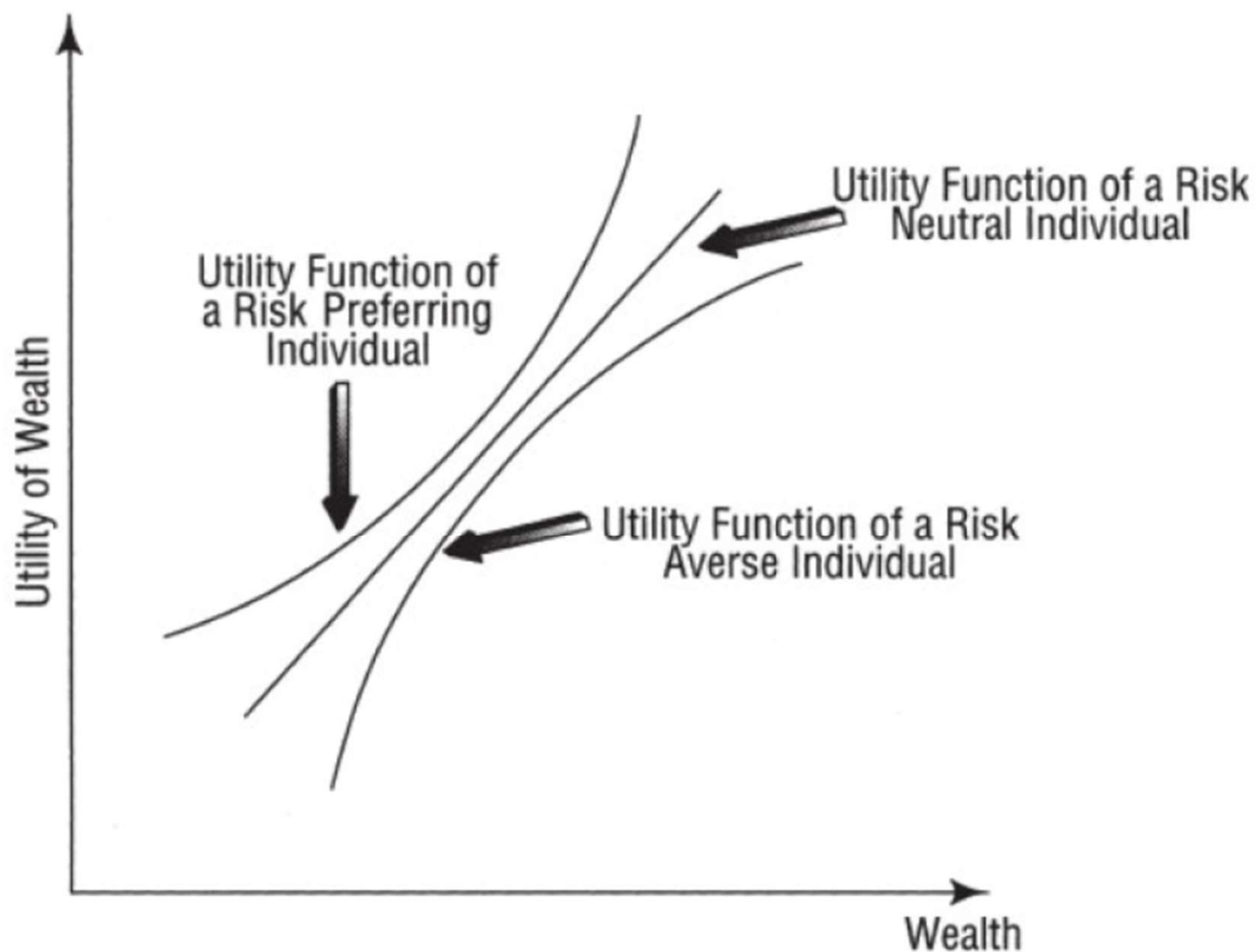
Risk Averse Person

- Assume a gamble with two possible outcomes for wealth W_1 and W_2 , each with 50% probabilities
- Then the expected wealth from this gamble is $E(W)=0.5*W_1+0.5*W_2$
- This will lie midway on the straight-line joining the points W_1 and W_2 on the curve
- Also, if one computes the expected utility of wealth $E[U(W)]$, the same will fall midway between $U(W_1)$ and $U(W_2)$
- However, as can be seen from the graph, to generate the same level of utility $E[U(W)]$, a lower level of certain (ensured) wealth is required

Risk Averse Person

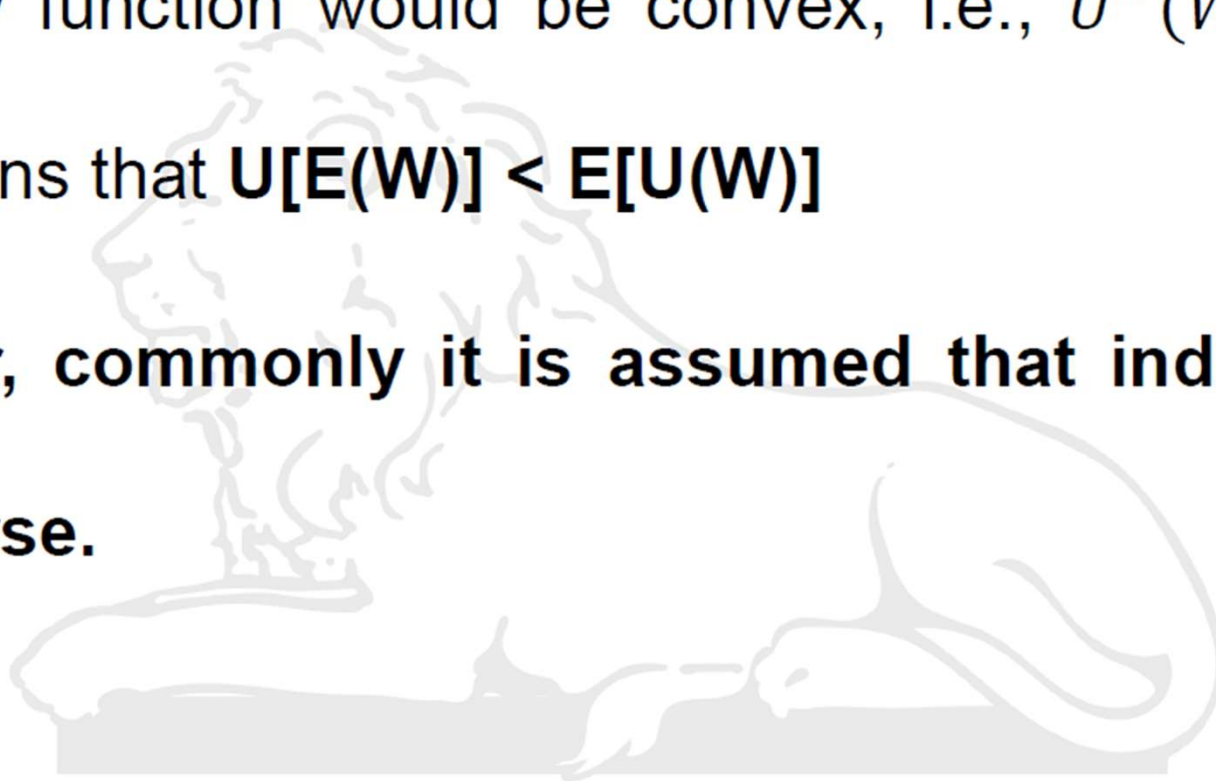
- It is very easy to observe here that, for this kind of utility function $U[E(W)] > E[U(W)]$
- The more concave this function (downward sloping) higher the difference between $U[E(W)]$ and $E[U(W)]$, i.e., the individual will be more risk averse
- A sure payment that makes this individual indifferent between gamble and sure payment is “Certainty Equivalent” (CE)
- As can be seen the certainty equivalent is less than $E(W)$ and this difference represent the risk-premia of this risk-averse individual

Introduction: Risk Preference

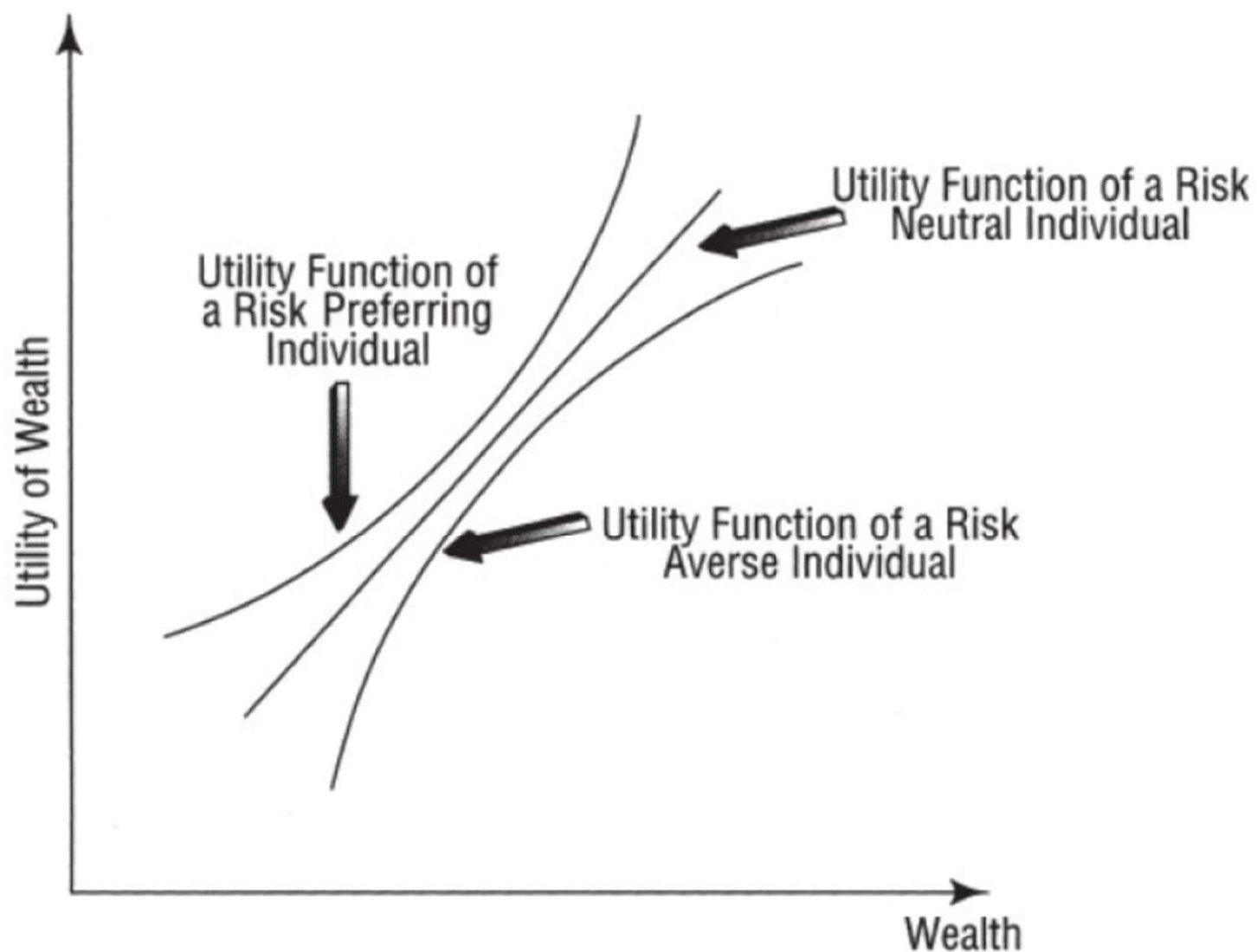


Risk Preferring Person

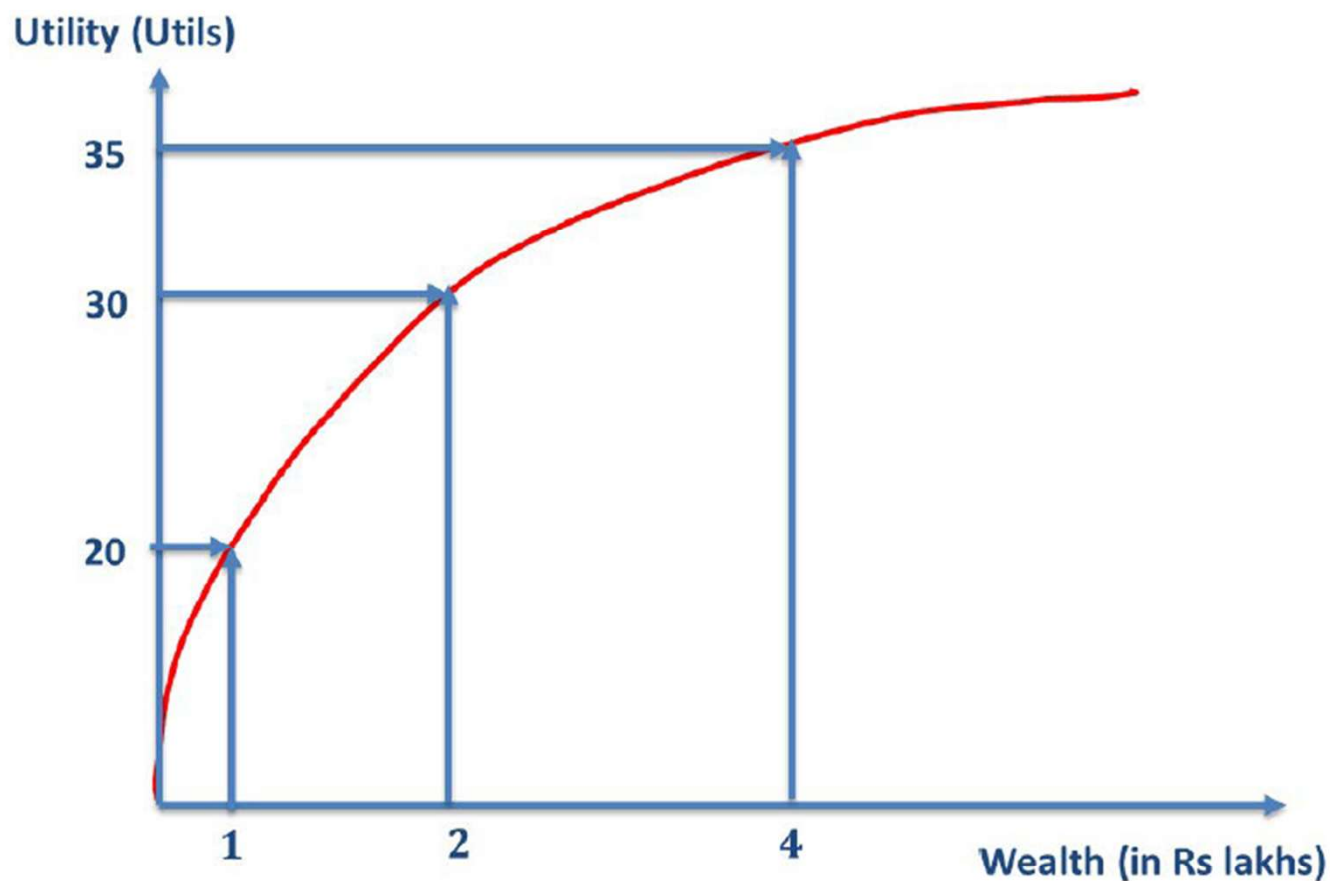
- A risk preferring individual would prefer the riskier situation
- His utility function would be convex, i.e., $U''(W) > 0$. This also means that **$U[E(W)] < E[U(W)]$**
- **However, commonly it is assumed that individuals are risk averse.**



Introduction: Risk Preference



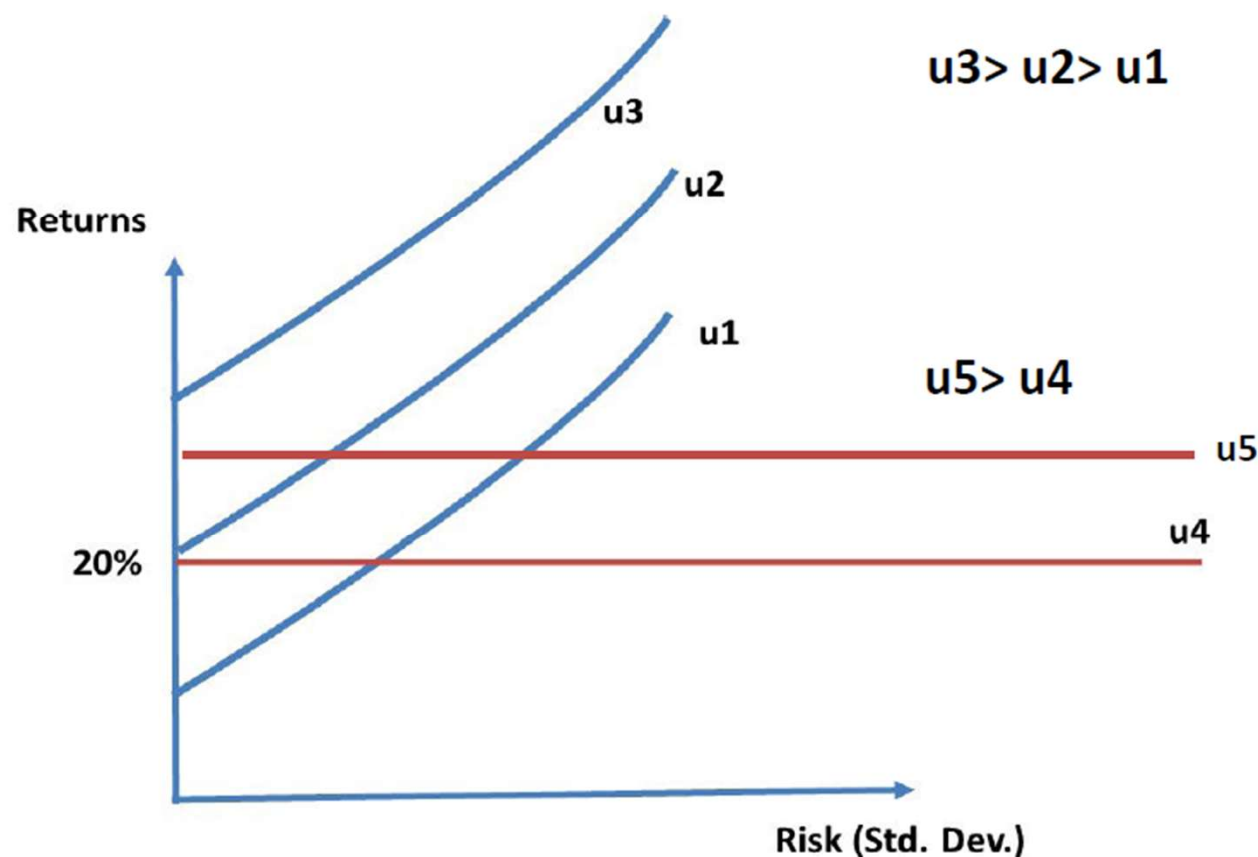
Wealth utility diagram for risk averse person



Wealth vs utility for risk averse person

- More wealth will always have more utility (Non-satiation)
- However, for a risk-averse investor, the relationship exhibits diminishing marginal utility
- For example, an increase of 1 Lakhs from 1 Lakhs to 2 Lakhs results in 10 unit of additional utility
- However, the next 2 Lakhs offer only 5 unit of additional utility
- So, the expected returns have to be increased to reach the same amount of additional utility.

Indifference curve for risk and Expected return (SD)

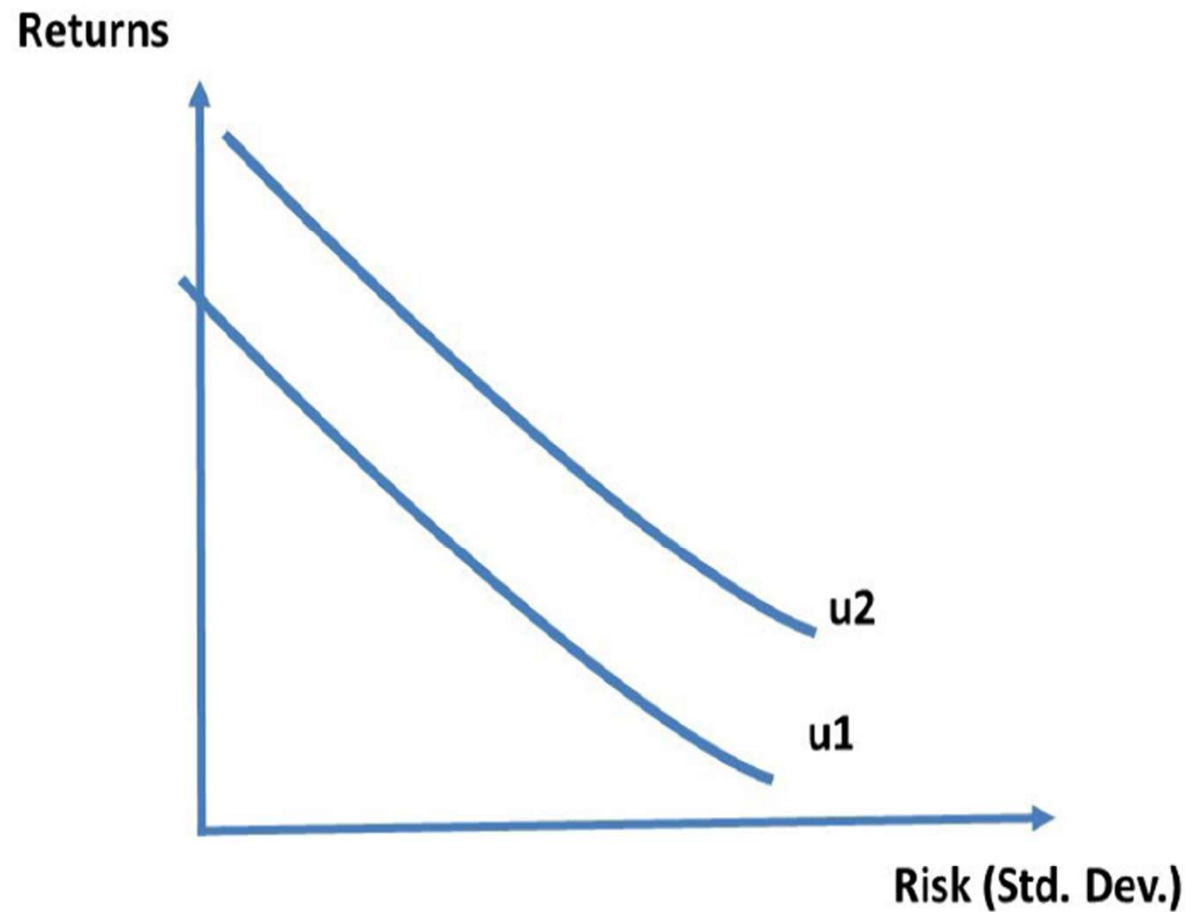


Indifference curve for risk and Expected return (SD)



- In this graph, $u_3 > u_2 > u_1$ are indifference curves for an individual
- This means increasing level of expected returns for the same risk or the same level of returns for lower risk lead to higher utility
- **What is the nature of this person?**
- However, u_4 is horizontal. For this individual, utility remains same despite the increase in risk
- That is the marginal utility does not diminish with increasing risk
- **What is the nature of this person?**

Return utility graph for increasing marginal utility



Return utility graph for Risk preferring person

- For this person, increase in risk results in lower level of return expectations. It means that he enjoys risk (gambler, speculator).



Thanks
